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Research article

Enhancing 2D/3D heterointerface via argon plasma polishing for efficient perovskite solar cells

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ABSTRACT

Low-dimensional perovskite top-surface passivation is crucial for improving the performance of n-i-p perovskite solar cells. However, the soft lattices present at the perovskite interface inhibits the effectiveness of 2D passivation. Here, we utilized argon plasma polishing pre-treatment (APP) to remove surface defective layer of the perovskite, thereby exposing solid sub-surface of perovskite films. Our findings revealed that APP can effectively remove impurities adsorbed on the surface of thin films, eliminate perovskite soft lattices. Furthermore, APP can adjust the surface composition of perovskite, which is beneficial for subsequent low-dimensional perovskite passivation. The transformation of organic amine salt ligands into 2D perovskite is more complete, resulting in a higher-quality 2D/3D heterointerface and a significant reduction in non-radiative recombination of charge carriers. The APP strategy contributes to the stability of the perovskite interface and successfully improves the average efficiency of perovskite solar cells from 22 % to 22.8 %, with a maximum efficiency exceeding 23.5 %.

1. Introduction

Perovskite solar cells (PSCs) are recognized as the most promising next-generation photovoltaic technology for commercialization, with power conversion efficiencies (PCEs) surpassing 26 % [1–4]. Nonetheless, complex surface defects in perovskite thin films still limit photovoltaic devices from achieving higher PCEs [5–7]. Passivation is essential for the perovskite surface, and the effectiveness of this passivation depends significantly on the original surface characteristics [8]. Further improvements in the stability and performance of PSCs require precise management of perovskite surfaces [9].

To date, numerous materials have been confirmed to effectively passivate surface and interface defects in perovskite solar cells [10–12]. For instance, Lewis bases can effectively passivate undercoordinated Pb ions on the surface of perovskite thin films, whereas Lewis acids can effectively passivate Lewis base traps [13–16]. Moreover, two-dimensional (2D) and one-dimensional (1D) perovskite, when used as an interfacial capping layer, not only effectively passivates perovskite surfaces but also forms a well-defined heterojunction structure, especially beneficial for n-i-p structured perovskite solar cells due to its band energy-level alignment [17–21]. Such as 2D $(BA)_2PbI_4$, $(TEA)_2PbX_4$ (X = I or Br), and 1D (TBA)PbI_3 and (TMS)PbI_3, when combined with 3D perovskite heterointerfaces, can assist in reducing defects and enhancing stability against humidity, thermal degradation, and operational stress [22–25].

However, perovskite films prepared by the solution method exhibit soft lattices on the surface, including nanocrystals and amorphous phase regions with numerous surface defects and closely packed grain boundaries, which adversely affect surface passivation [6,26,27]. Therefore, it is essential to pretreat the perovskite films prior to passivation to ensure more robust results. Chemical solvent polishing, using solvents such as acetonitrile (ACN), 2,2,2-trifluoroethanol, and methanol, can dissolve the organic iodide in the perovskite layer, resulting in a PbI_2-rich surface that provides reactive sites [28–30]. Additionally, additive engineering in the carrier transport layer, such as with Spiro-OMeTAD precursor, can also polish the surface of perovskite

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[31,32]. The creation of 2D perovskite to eliminate undesired metastable amorphous species and residual PbI_2 nanocrystals, followed by the removal of the 2D perovskite and excess polishing agents, effectively passivates the interface [33]. Besides, mechanical cleavage using chemically modified adhesive tape peel-zone and physical polishing have proven effective for removing surface defective layer of perovskite [34,35]. Despite this, these methods lack precise controllability, uniformity, reproducibility, and are not suitable for large-scale application. The time window for ligands to transform into 2D perovskite is limited, highlighting the need to develop state-of-the-art technology to enhance the formation of 2D/3D perovskite heterojunctions.

In this work, we report an argon plasma polishing pre-treatment (APP) as an effective method to remove the surface defective layer of the perovskite, thereby enhancing the passivation effectiveness by 2D/3D perovskite heterointerface. We revealed the presence of soft defective layer in perovskite films through two types of etching rates. And we found that APP can effectively remove impurities adsorbed on the surface of thin films, eliminate perovskite soft lattices and enhance the extraction and transport of charge carriers. Furthermore, APP can adjust the surface composition of perovskite, which is beneficial for subsequent low dimensional perovskite passivation and reduces non radiative recombination of charge carriers. The transformation of passivating ligands into 2D perovskite is more complete, resulting in a higher-quality 2D/3D heterointerface. The APP strategy contributes to the stability of the perovskite interface and successfully improves the average efficiency of perovskite solar cells from 22 % to 22.8 %, with a maximum efficiency exceeding 23.5 %.

2. Experimental section

2.1. Materials

N, N-dimethylformamide (DMF, 99.8 %), dimethyl sulfoxide (DMSO, ≥99.9 %), chlorobenzene (CB, 99.9 %) were purchased from Alfa Aesar. Formamidinium iodide (FAI, 99.5 %), methylamine hydrobromide (MABr, 99.5 %), methylamine hydrochloride (MACl, 99.5 %), Phenethylamine Hydroiodide (PEAI, ≥99.5 %), lead bromide ($PbBr_2$, 99.9 %) were bought from Xi'an Polymer Light Technology. Lead iodide (PbI_2, >98 %) was received from TCI. $SnCl_2{\cdot}2\,H_2O$ (99.99 %), thioglycolic acid (TGA, 98 %) were received from Aladdin. Urea (>99 %), 4-tert-butylpyridine (TBP, 98 %), Potassium Chloride (KCl, 99.0 %), Octylammonium Iodide (OAI, 99.0 %) and lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI, ≥99.9 %) was obtained from Sigma-Aldrich. 2,2′,7,7′-tetrakis (N, N-dip-methoxyphenylamine)-9,9′-spirobifluorene (Spiro-OMeTAD, >99.8 %) was purchased by Luminescence Technology Corp. Isopropanol (IPA, 99 %) was bought from Macklin. Gold (99.999 %) was purchased from Zhongnuo Advanced Material (Beijing) Technology Co., Ltd.

2.2. Device fabrication

FTO glass substrates were ultrasonically cleaned in deionized water, acetone, and ethanol for 20 min each. Subsequently, a SnO_2 film was deposited onto the FTO glass using the chemical bath deposition (CBD) method. The CBD solution was prepared by dissolving 275 mg of $SnCl_2{\cdot}2H_2O$, 1250 mg of urea, 25 μL of TGA, and 1.25 mL of hydrochloric acid in 100 mL of deionized water. The solution was reacted in a water bath at 94 °C for 5 h and 40 min. After the reaction, the FTO glass was cleaned with dilute hydrochloric acid and isopropanol solutions by ultrasonic cleaning for 5 min. The cleaned FTO glass was then dried with nitrogen gas and annealed on a hot plate at 170 °C for 1 h. Next, a 10 mmol potassium chloride solution was spin-coated onto the substrate at 3000 rpm for 30 s, with an acceleration to 1000 rpm, and then annealed at 100 °C for 10 min. Prior to transfer to the glove box, the substrate was treated with ultraviolet ozone for 15 min. A perovskite precursor solution of 1.5 M $(FAPbI_3)_{0.94}(MAPbBr_3)_{0.06}$ was prepared with a 4 % excess of PbI_2 and 0.5 M MACl, dissolved in 115 μL of DMSO and 885 μL of DMF. This perovskite precursor was spin-coated at 1000 rpm for 10 seconds (200 rpm ramp) and 5000 rpm for 30 s (1000 rpm ramp), with 120 μL of antisolvent (CB) dropped at 15 s during the 5000 rpm spin. The perovskite films were then annealed at 120 °C for 15 min in air (25 % relative humidity). After annealing, perovskite films were treated with or without APP. Following this, 10 mmol PEAI or OAI dissolved in isopropanol was spin-coated at 3000 rpm for 30 s and annealed at 60 °C for 5 min. Afterward, 20 μL of Spiro-OMeTAD solution was spin-coated onto the perovskite films at 3500 rpm for 30 s and annealed at 60 °C for 5 min. The Spiro-OMeTAD solution was prepared by dissolving 85.78 mg of Spiro-OMeTAD, 39 μL of TBP, 35 μL of Li-TFSI solution (520 mg mL^{-1} in acetonitrile), and 12 μL of Co-TFSI solution (300 mg mL^{-1} in acetonitrile) in 1 mL of CB. Finally, an 80 nm gold electrode was deposited onto the films by thermal evaporation.

2.3. Argon plasma treatment

Argon plasma treatment uses SY-DT02S from Opps Plasma Technology Co., Ltd. Place the prepared perovskite thin film in a plasma processing instrument, set the power from 100 to 400 W, and the processing time depends on the experimental content. The gas flow was set to 100 SCCM, the vacuum level was maintained at 20 Pa, and the chamber size was 260 mm × 260 mm × 260 mm (17 L).

2.4. Device characterizations

Current density–voltage (J–V) curves were measured using a solar simulator (Newport, Oriel Sol 3 A) and a digital source meter (Keithley 2450). Light intensity was set to AM 1.5 G and calibrated to 100 mW cm^{-2} using a calibrated silicon solar cell. The step voltage was 10 mV and the delay time was 50 ms. A metal mask with an area of 0.09 cm^2 was applied to the top of the cell. The XPS was measured using a Thermo Scientific ESCALAB 250Xi spectrometer with Al Kα (1486.6 eV) radiation and the argon plasma bombardment within the XPS instrument is 2000 eV. PL and TRPL were carried out by FS5 (Edinburgh) with the excitation at 475 nm and 405 nm. SEM images were obtained by a field-emission SEM (JSM-7800F). XRD patterns were acquired by Mini Flex II at a scan rate of 10° min^{-1}. AFM images were obtained by Dimension Icon (Bruker). UV–vis spectra were obtained on a UV–vis spectrometer (SR-500I-A).

3. Results and discussion

Plasma cleaning is a common method for cleaning organic contaminants from material surfaces using energetic argon plasma [36]. Firstly, we analyzed the changes in O 1 s orbitals and C 1 s orbitals on the surface of perovskite thin films using X-ray photoelectron spectroscopy (XPS), as shown in Fig. 1a-b. The APP process was carried out in-situ with argon ion sputtering attachment of XPS. And the full in-situ XPS spectra of perovskite films with different APP time are depicted in Figure S1. It is evident that oxygen is present on the surface of the perovskite film. The signals of oxygen elements could be divided into two categories, one belonged to metal carbonate, and another belonged to metal hydroxide [37]. Oxygen is always present on the surface of samples exposed to air due to external contamination, oxidation, or the presence of water. When the surface of the perovskite film is treated with argon plasma, the signal of oxygen element disappeared as shown from Fig. 1a. Additionally, a similar effect was observed in the C 1 s signal, as shown in Fig. 1b. Without APP, there were three distinct C 1 s signals on the perovskite surface. The strongest signal at 284.8 eV originated from contaminant carbon. The weakest signal at 286.5 eV was attributed to the C-O single bond, and the signal at 288.2 eV belonged to the cation in the perovskite lattice [38]. As the time of APP increased, the intensity of the 284.8 eV peak, originally the strongest, decreased significantly, and the C-O single bond signal disappeared. Only the lattice carbon signal

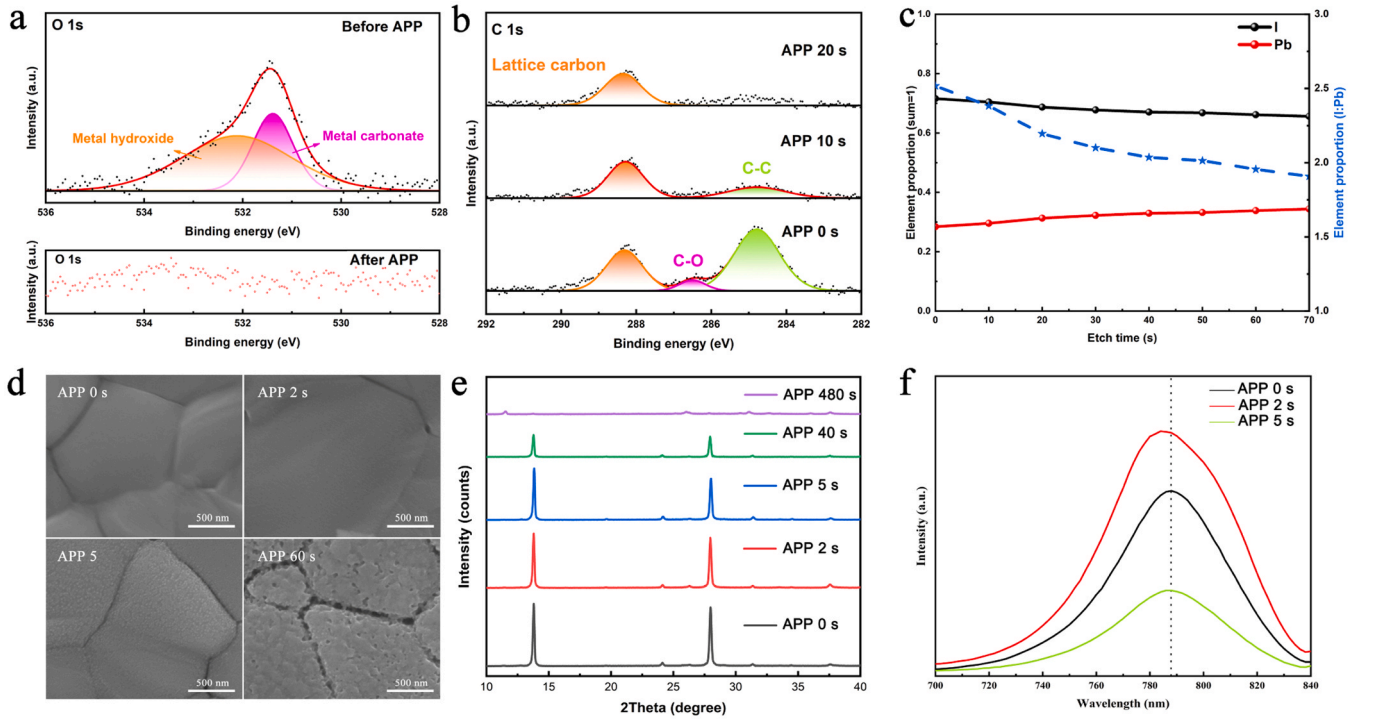


Fig. 1. a) XPS spectra of O 1 s and b) C 1 s of the perovskite film with different polishing time. c) Element ratio of perovskite films with different polishing time (10–70 s). d) Top-view SEM images of perovskite films with different APP times. e) XRD results of perovskite films with different APP times. f) Steady state PL spectra of perovskite films with APP for different time (0, 2 and 5 s). In addition to XPS testing, the APP power was optimally set to 200 W.

remained almost unchanged. This indicates that the contaminant carbon and impurities adsorbed on the perovskite film surface were effectively removed by APP.

We further assessed the surface composition changes of perovskite films with APP. By comparing the changes in lead and iodine content, the iodine-to-lead atomic ratio on the surface was calculated, as shown in Fig. 1c. It was observed that APP gradually increased the proportion of lead on the surface of the perovskite film, corresponding to a decrease in the proportion of iodine. This indicates that argon plasma reconstructs the surface interface of the perovskite, removing some iodine while leaving the lead on the surface. The iodine-to-lead atomic ratio on the perovskite surface decreased from 2.5:1 initially to 2.0:1 after 60 s, suggesting a lead-rich perovskite surface. This indicates that organic formamidine iodide (FAI) is easily removed from the surface. Additionally, with extended APP time, the signal of metallic Pb signal becomes significant, as shown in Figure S2 [39]. Moreover, the surface morphology changes under different APP times, as seen from the scanning electron microscopy (SEM) images (Fig. 1e and Figure S3). After 2 seconds of argon plasma treatment, the surface appeared relatively intact. However, with extended treatment time, the perovskite surface was etched with a large number of small particles on the perovskite surface, followed by the formation of numerous pinholes. Prolonged APP can be detrimental to organic-inorganic lead halide perovskite materials. As shown in Fig. 1e, X-ray diffraction (XRD) results indicated that the intensity of the perovskite diffraction peak at around 14° significantly decreased when APP was extended to 5 seconds. Furthermore, when the APP time reached 480 seconds, the signal of the perovskite phase disappeared and was replaced by signals from other compounds. The UV–vis absorption spectra of the perovskite film, shown in the supplementary information (Figure S4), also demonstrated that prolonged APP leads to perovskite decomposition. Steady-state photoluminescence spectra showed that APP treatment for 2 seconds exhibited the strongest fluorescence intensity and a slight blue-shift, indicating that transient APP treatment is beneficial for the elimination of surface defects and suppression of the band-tail state (Fig. 1f) [40]. The PL decay lifetimes (τ) were fitted using a mono-exponential model, as shown in Figure S5 and Table S1, with τ values of 1615 ns, 1812 ns, and 1138 ns for the control, APP 2 s, and APP 5 s, respectively. These results demonstrate that transient APP treatment can slightly increase the carrier lifetime. Therefore, these findings confirm a strong interaction of APP on the perovskite surface.

To further confirm the effect of APP on the crystalline structure, a series of cross-sectional SEM images of perovskite thin films with APP for different durations were analyzed, as shown in Fig. 2a and Figure S6. The thicknesses of the perovskite layers were approximately 810 nm, 780 nm, 740 nm, 710 nm, 680 nm, 640 nm, and 610 nm for APP durations of 0 s, 2 s, 5 s, 10 s, 20 s, 40 s, and 50 s, respectively. We further analyzed these data and plotted them as a scatter plot, as shown in Fig. 2b. Fitting the data revealed two-step linear correlations between the thickness of the perovskite film and the APP processing time. The initial etch rate was 14 nm/s, followed by a subsequent etch rate of 2.5 nm/s. This indicates that the perovskite thin films indeed have a surface defective layer. The soft lattice is weak and has more lattice defects, making it more susceptible to decomposition by argon plasma. Atomic force microscopy (AFM) results showed a slight decrease in the arithmetic average roughness (RA) with increased APP treatment time, as shown in Figure S7. Considering that APP's etching effect would typically increase surface roughness, this also indicates the presence of numerous defect states on the perovskite surface. When the APP treatment time is extended to 60 s, cracks are observed, and the roughness average (RA) significantly increases to 32.2 nm (Figure S8), indicating that prolonged plasma treatment causes severe damage to the grain boundaries, which is consistent with the SEM results.

Furthermore, we carried out grazing incidence angle X-ray diffraction (GIXRD) to explore the lattice distortion at different depths of perovskite films and the results were shown in Fig. 2c and d. Using the (100) crystal plane as a reference, the surface lattice spacing was found to be smaller than that of the bulk lattice. For the untreated perovskite films, the absolute value of the percentage change is 0.636 %, while for films of APP 2 s, it is 0.497 % for grazing incidence angle from 1° to 2°.

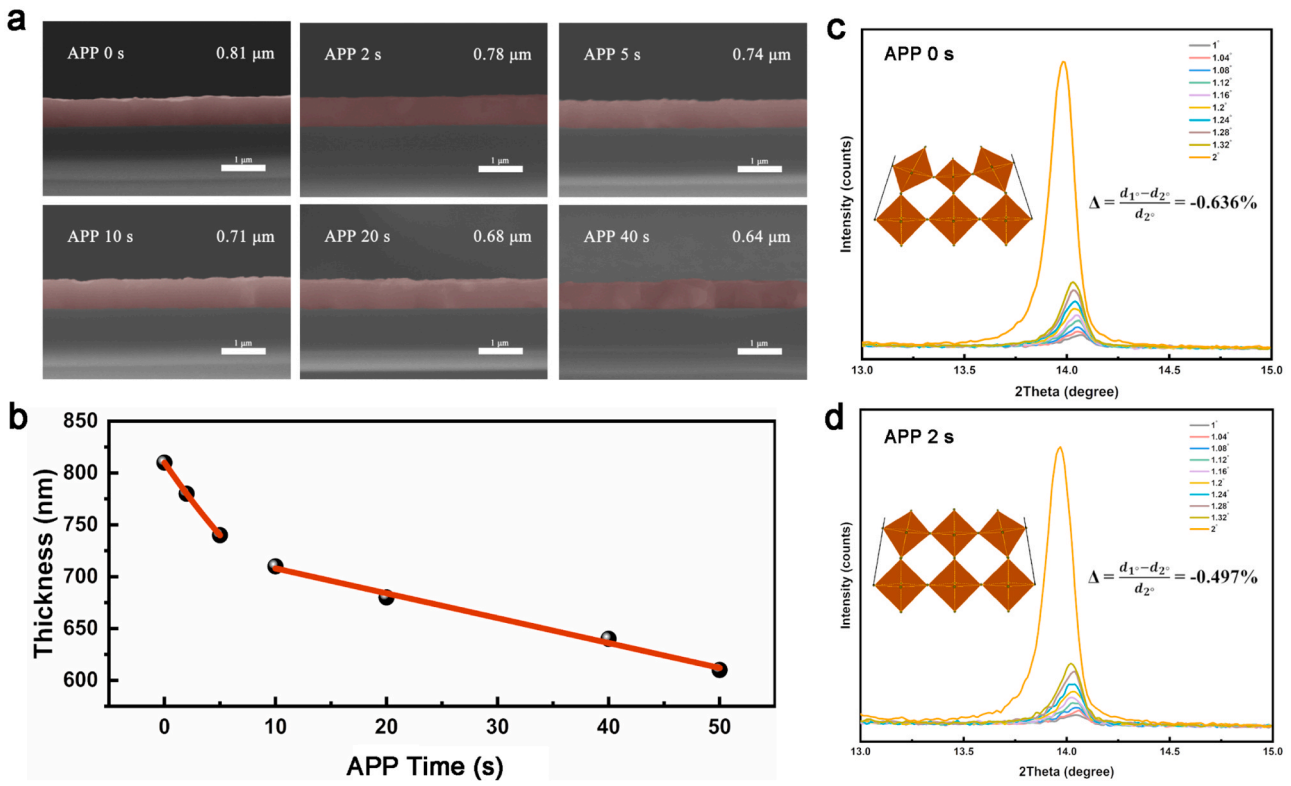


Fig. 2. a) Cross-sectional SEM images of perovskite films with different APP time (0, 2, 5, 10, 20 and 40 s) and b) thickness of perovskite layer with different APP time. c) and d) GIXRD of perovskite films with different grazing angle without (APP 0 s) and with APP 2 s, respectively.

The difference in lattice spacing decreased after APP treatment. This indicates that the untreated perovskite thin films have a highly deformed crystal structure on the surface, while APP effectively removes these deformed crystal lattices. In addition, we have conducted contact angle measurements to evaluate the surface energy of the perovskite surface wi/wo APP using the Owens-Wendt-Rabel-Kaelble (OWRK)

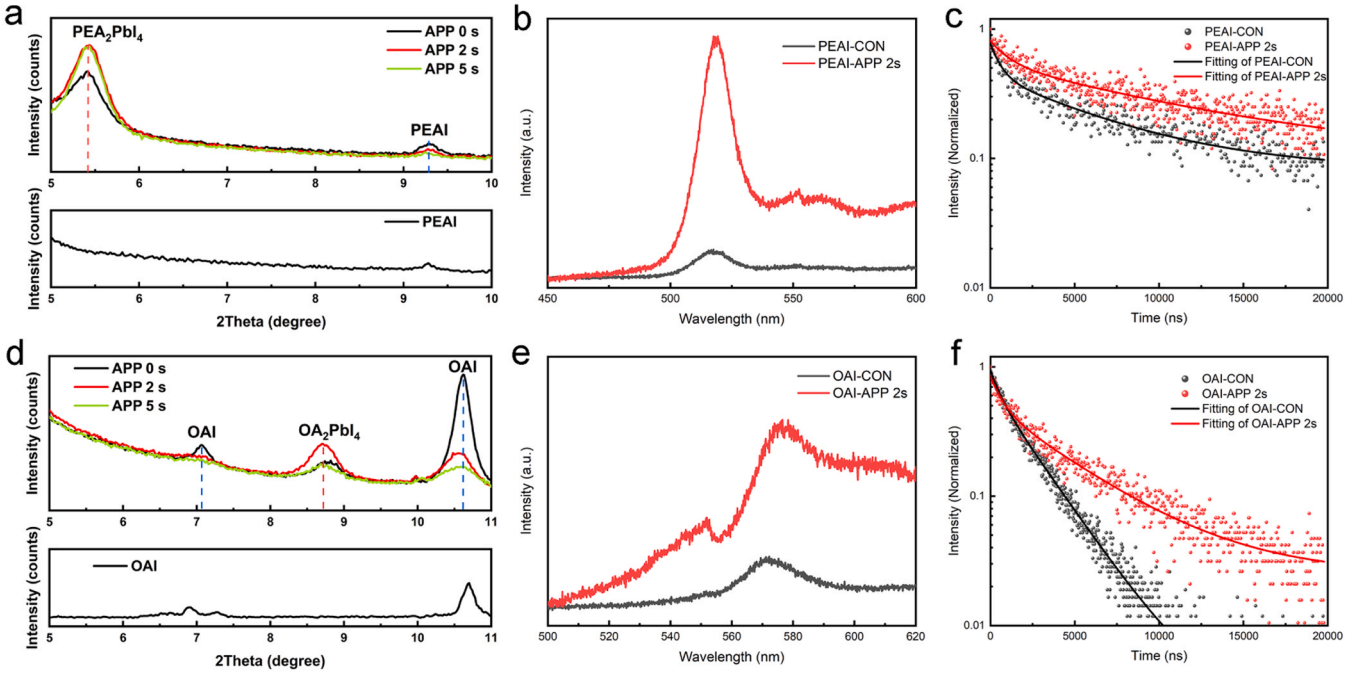


Fig. 3. a) XRD patterns of PEAI post-treatment passivation with different APP times (0, 2 and 5 s). b) PL spectra of 2D perovskite (PEA_2PbI_4) form on bulk-perovskite without and with APP treatment (control 0 s and APP 2 s). c) TRPL spectra of PEAI passivated perovskite film (emission 788 nm) without and with APP treatment (control 0 s and APP 2 s). d) XRD patterns of OAI post-treated perovskite films with different APP time (0, 2, and 5 s). e) PL spectra of 2D-perovskite (OA_2PbI_4) form on bulk-perovskite without and with APP treatment (control 0 s and APP 2 s). f) TRPL spectra of OAI passivated perovskite film (emission 788 nm) without and with APP treatment (control 0 s and APP 2 s). Note: XRD patterns of PEAI and OAI are shown as a reference.

method. As shown in Figure S9 and Figure S10, the calculated surface energies of the perovskite surface with APP for 0 s, 2 s, and 5 s are 44.85 mN/m, 45.54 mN/m, and 46.33 mN/m, respectively. The change in surface energy is negligible, the slight increase in surface energy may contribute marginally to the subsequent organic amine salt transformation.

For n-i-p structured PSCs, 2D passivation applied to the top surface of the perovskite layer is effective in improving performance. We then compared the effects of 2D passivation on the APP-treated perovskite surface. Phenethylamine hydroiodide (PEAI) with an aromatic ring and octyl ammonium iodide (OAI) with an aliphatic chain are commonly used organic amine salts for passivating perovskite surfaces [41,42]. We treated the perovskite surface with PEAI and OAI to form a passivation layer with or without APP. The XRD results are shown in Fig. 3a and d, and the typical full XRD patterns are presented in Figures S11–S12. For organic salts simply depositing on the surface of the perovskite layer, the 2D transformation is uncompleted with residual organic matters. While for perovskite sub-surface by APP, the 2D transformation is completed, indicated by the stronger diffraction peak signals for both PEA_2PbI_4 and OA_2PbI_4 passivation.

To further study the optical characteristics of the perovskite films after 2D passivation, we utilized a 405 nm laser to explore the subsurface properties for the 2D passivation effect. The PL spectra indicate that the 2D perovskite PEA_2PbI_4 shows stronger emission at 517 nm with transient APP compared to the control group, as shown in Fig. 3b. In addition, time-resolved photoluminescence (TRPL) spectra were obtained to assess the photogenerated carrier lifetime using a bi-exponential model. An ultralong average carrier lifetime of 7218 ns was achieved with PEAI-APP, more than twofold longer than the 3528 ns observed with PEAI-Control, as depicted in Fig. 3c and Table S2. The OAI-treated perovskite interfaces, with APP treatment, also showed stronger 2D PL intensity and longer overall carrier lifetimes compared to the control group without APP treatment (Fig. 3e, f, and Table S3). This result aligns with the findings for the PEAI-treated perovskite interfaces, underscoring the improvement in heterointerface quality with APP treatment.

Based on the aforementioned results, we developed a schematic for 2D ligand passivation of perovskite films, as shown in Fig. 4. In the conventional approach, the amine ligand only partially forms a 2D capping layer, leaving some unreacted amine ligand on the surface and resulting in incomplete passivation with residual non-radiative recombination. In contrast, APP enhances the transformation of organic ammonium salts into 2D perovskite and significantly improves passivation by effectively suppressing non-radiative recombination from surface defect traps. The APP strategy effectively accelerates the transformation capability of 2D/3D heterointerfaces and has significant potential for surface passivation in large-area perovskite devices.

We further evaluated the overall device performance. Space charge limited current (SCLC) measurements were used to quantify the trap density (N_t) of perovskite devices with the structure FTO/SnO_2/perovskite (APP or not)/2D/PCBM/Au, designed for electron transport only. As depicted in Fig. 5a and b, the trap-filled limited voltage (V_{TFL}) is 0.64 V for the Control and 0.53 V for the devices with APP, corresponding to the N_t of 3.45×10^{15} cm^{-3} of Control to 3.08×10^{15} cm^{-3} for the devices with APP, respectively [43]. These findings are consistent with the PL and TRPL results. The analysis of electrochemical impedance spectroscopy (EIS) demonstrates that the APP-treated perovskite surface effectively promotes the transformation of organic amine salts into a 2D perovskite structure, forming a better 2D/3D heterointerface, as shown in Figure S13. This, in turn, effectively enhances the recombination resistance (R_{rec}) and reduces interfacial carrier recombination.

We fabricated the devices with the architecture of FTO glass/SnO_2/perovskite/2D/Spiro-OMeTAD/Au with adjustable APP, where Spiro-OMeTAD is 2,2',7,7'-tetrakis (N, N-dip-methoxyphenylamine)-9,9'-spirobifluorene. The discharge power of the argon plasma was optimized from 100 to 400 W, and the best-performing devices were treated at 200 W, as shown in Figure S14. Fig. 5c presents the box plot of PCEs with APP for different times (0, 2, and 5 seconds) under the passivation of PEAI and OAI. The detailed parameters of short-circuit current density (J_{SC}), open-circuit voltage (V_{OC}), and fill factor (FF) are shown in Figure S15. It is apparent that the PSCs treated with argon plasma for 2 seconds had the highest average PCEs. The average efficiency of perovskite solar cells increased from 22 % in the control to 22.8 % for PEAI-APP, and from 21.6 % in the control to 22.5 % for OAI-APP. The typical J-V curves of photovoltaic devices are shown in Fig. 5d and e. The champion PCE of 23.53 % was achieved with PEAI-APP 2 s, with a J_{SC} of 24.45 mA/cm^2, V_{OC} of 1.18 V, and FF of 81.5 %. The external quantum efficiency (EQE) spectra of the APP 2 s PEAI-passivated photodevice shows an improvement in current integration, with a value of 24.36 mA/cm^2 for APP 2 s, as shown in Figure S16. The OAI-APP 2 s also showed a higher PCE of 23.10 % than the control. The performance improvement is mainly attributed to the increase in V_{OC}, indicating a better passivation effect by exposing the solid perovskite sub-surface for 2D transformation and effectively suppressing non-radiative recombination. A slight increase in J_{SC} suggests improved carrier extraction and transport at the perovskite interface. However, 5 seconds of APP significantly reduces J_{SC}, indicating that excessive treatment can detrimentally affect the perovskite sub-surface by the formation of numerous pinholes. Furthermore, the ISOS-Dark-Inert stability tests were conducted, as shown in Fig. 5 f. The device without encapsulation maintained over 80 % of its initial efficiency by the fourth week with APP treatment for 2 seconds, while the efficiency of the control group decreased rapidly to 73 %. The ISOS-Light-1I stability tests also demonstrate better stability with 2 seconds of APP treatment, as shown in Fig. 5 g. The improved stability indicates that further 2D passivation, achieved by removing the perovskite soft lattice, contributes to better heterointerface properties.

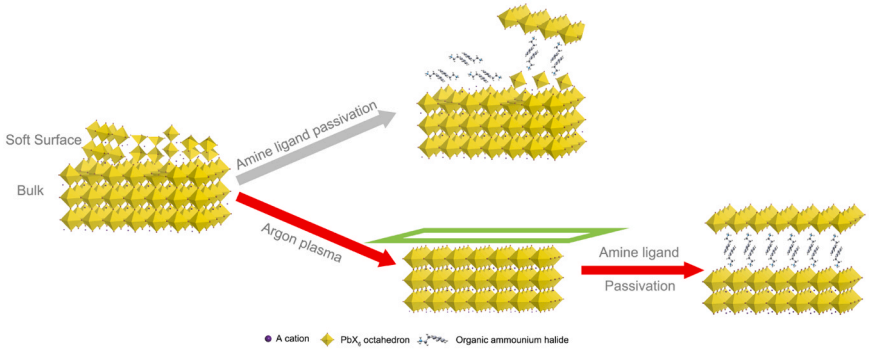


Fig. 4. A schematic illustrating the surface passivation of perovskite layers with and without APP.

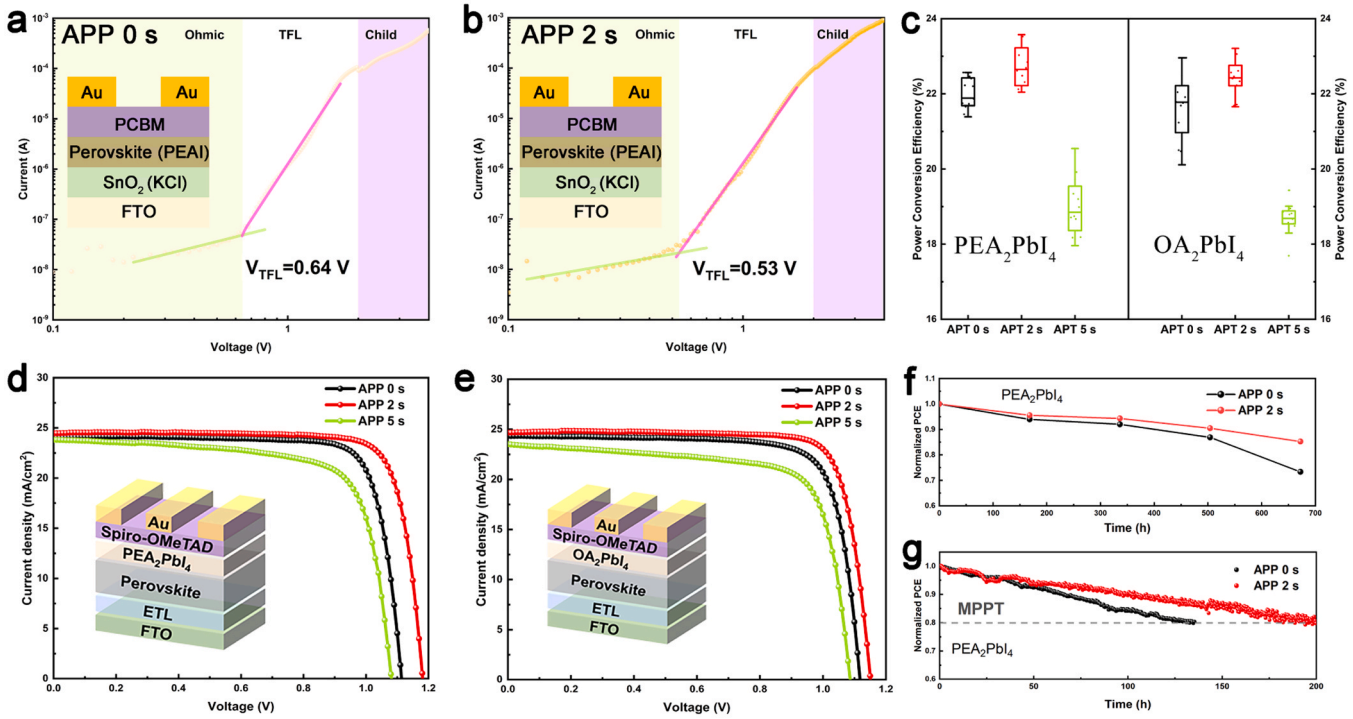


Fig. 5. a) and b) I-V curves of space charge limited current curves (SCLC) measurement of devices with FTO/SnO_2/perovskite/PEAI/PCBM/Au structure without and with APP, respectively. c) Box plot of PCEs with APP for different time (0, 2 and 5 s) under the passivation of PEAI and OAI, based on statistics from over 40 devices. d) Photovoltaic performance of the planar PSCs with APP for different time (0, 2 and 5 s) under the passivation of PEAI and e) OAI. f) Devise stability of APP 0 and 2 second storing in N_2 glove box under dark condition. g) Devise stability of APP 0 and 2 second under maximum power point tracking (MPPT) with 1-sun LED simulation.

4. Conclusion

In summary, we introduced APP into the manufacturing process of perovskite solar cells, effectively reconstructing the surface interface of perovskite films by removing contaminants and the soft defective layer from the perovskite surface, thereby exposing a solid sub-surface rich in the Pb-I framework. This method successfully improved the PCEs of perovskite solar cells by generating stronger 2D/3D perovskite hetero-interface, demonstrating consistent effects with different passivation organic ammonium salts. Additionally, APP contributed to enhancing the stability of the devices, offering valuable insights for optimizing surface passivation techniques and improving the commercial viability of perovskite solar cells.

CRediT authorship contribution statement

Hongshi Li: Writing – original draft, Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization, Project administration and Funding acquisition. **Tiantian Lou:** Formal analysis, Methodology and Data curation. **Jianqiang Wu**: Writing – original draft, Methodology, Formal analysis, Investigation and Data curation. **Guichun Yang:** Formal analysis. **Peng Chen:** Formal analysis and Funding acquisition. **Guoran Li:** Supervision, Conceptualization, Writing – review & editing and Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nxmate.2025.100779.

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